Startup Guide

FANUC CR-4iA, CR-7iA, CR-7iA/L, CR-14iA/L

Thank you for purchasing a FANUC Robot

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

Required Tools

- Straight-head screwdriver
- Cross-head screwdriver
- Torque wrench
- Metric hex-head wrenches

Required Items

- Properly rated transport equipment as specified
- Customer-supplied properly rated power source as specified on the Controller Power Source Label
- Customer-supplied properly rated power input cable; conductor size of AWG14 to AWG10; terminal size is M5
- Multimeter
- Rigid platform to hold robot
- 6-M12 bolts
- Loctite #243
- 6-M12 flat washers

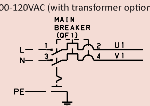
Optional Items

- Robot locator pins

Single Phase

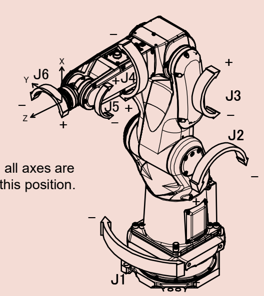
200-230VAC

100-120VAC (with transformer option)



Controller

Robot (mechanical unit) with joints indicated



Note: all axes are 0° in this position.

CR-7*i*A variant shown

A. Unpack and Intall your Robot

START HERE

1. After removing the shipping box, exposing the robot, remove the plastic covering from the robot.

Removing Shipping Box

Squeeze Tabs and Pull latches

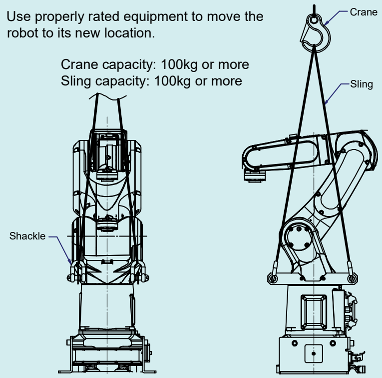
Shipment Skid

Robot, Controller Box, & Accessories Box

2. Remove the bolts holding the transport brackets to the skid.

3. Use properly rated equipment to move the robot to its new location.

Crane capacity: 100kg or more
Sling capacity: 100kg or more

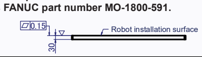


⚠ WARNING

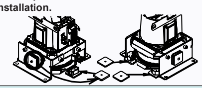
Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

Do not expose the robot to shock or exessive forces during installation or the internal sensor might break.

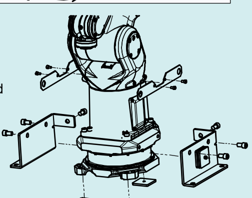
It is extremely important that the installation plate is flat after securing to the floor. Ensure there is no gapping between the installation plate and the floor; also confirm no debris is between the mounting surfaces. Fill any open spaces or gaps, especially directly underneath the robot base, with grout or similar method, so that the plate does not flex or become warped and so no gaps are present after installation. The installation surface should not be painted or fabricated. The installation plate is FANUC part number MO-1800-591.



Install spacers beneath robot sensor base for installation.

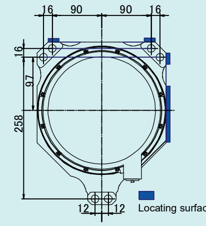


4. With the robot resting on the three spacers, remove the transportation brackets and hardware. Store these items and the robot skid in a safe, known location. These items are needed for future transportation.



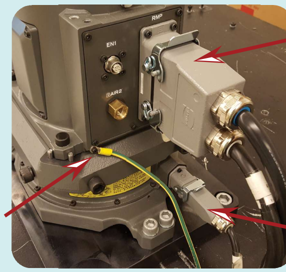
5. Gently raise the robot, remove the spacers, and again gently place the robot on to the installation plate.

6. Fasten the robot to the base plate with 6-M12 washers and 6-M12 bolts (tensile strength 1200N/mm² or more). Tighten them to a torque spec of 94Nm. If the highest accuracy is desired, locator pins can be used to ensure the robot and a future replacement are mounted in the same location. More information is available in the *CR-7iA Mechanical Unit Opertor's Manual.*



7. At the grounding location shown below, connect the ground wire from the controller to the robot base.

8. Connect the RMP cable and sensor cable as shown below.



9. Connect the teach pendant to its cable as shown below.

Rotate the teach pendant cable until it is seated. Then, spin the end of the connector until it is threaded on tight. Wiggling the connector might let you tighten it more.

Robot installation is complete. You are now ready to go to **B. Unpack and Install your Controller**.

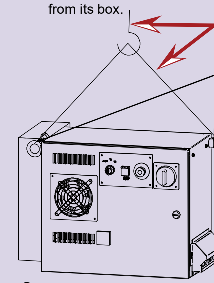
B. Unpack and Install your Controller

1. Open all of the boxes included in your shipment.

Store the items to the righgt in a safe, known location.

Hardware Bag, Sensor Calibration CD, Teach Pendant, & Inspection Data Sheet

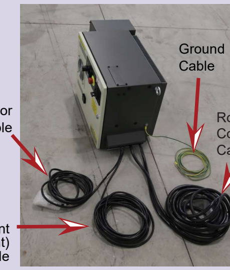
2. Use properly rated equipment to remove the controller from its box.



⚠ WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

3. Place the controller in a safe location outside of the robot work area such that the grounding wire, sensor cable, and robot connection cables can reach the robot. Leave enough space to open the controller door.

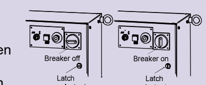


4. Find the power requirements for your controller on the lower right corner of the controller door.

⚠ WARNING

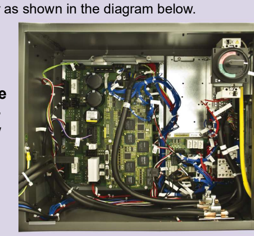
Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

5. Unlock the controller door by using a straight-head screwdriver to turn the silver latch counter-clockwise. To open the door, turn the Breaker counter-clockwise past its off position.

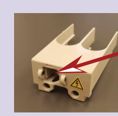


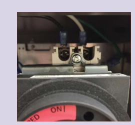
6. Route the user-supplied input power cable through the side of the controller as shown in the diagram below.

R-30*i*B Mate Controller - Inside View



7. To connect power, remove the plastic cover from above the Breaker (see the diagram below). To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.





Go to Page 2

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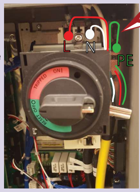
8. Connect power to the Breaker with the user-supplied input power cable. For single phase, connect L to 1 and N to 3. For three phase, connect L1 to 1, L2 to 3, and L3 to 5. Connect the ground to the plate located to the right of the Breaker.



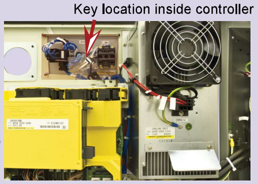
Breaker connection with custom-er-supplied input power cable

Breaker with cover reinserted

9. After the connections are made, reinsert the plastic cover.

10. Use a multimeter to verify the correct voltage is present.

Key location inside controller



11. Before you close the controller door, cut loose the set of keys from inside the door, and store them in a known location outside of the controller for use later in this guide.

12. Close the controller door and use a straight-head screwdriver to turn the silver latch on the outside of the controller lockwise.

Controller installation is complete. You are now ready to go to **C Turn on Your Controller**.

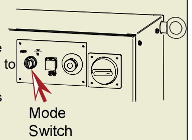
C. Turn on your Controller

⚠ WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

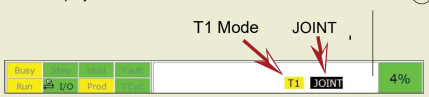
1. Turn on the controller by rotating the Breaker clockwise to ON.

2. Locate the Mode switch on the Controller Operator Panel. Use the key retrieved in Step B11 to switch to T1 mode. T1 mode limits the robot speed to 250 mm/sec and requires robot control to be at the teach pendant only.



3. Verify the controller is in T1 Mode by looking at the teach pendant screen. The mode indicator will be hilighted yellow and look similar to the status bar below. If it is not T1, repeat the previous step.

4. Press the COORD key on the teach pendant until JOINT is displayed similar to the status bar below.



Teach Pendant Status Bar

You have turned on your controller. You are now ready to go to **D. Jog your Robot**.

D. Jog your robot

1. Collaborative robot settings

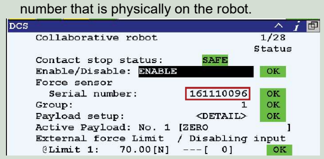
NOTE: To continue setup, the controller needs to have been powered on for 30 minutes or more. After this warmup time, continue below.

a. Using a computer, load the .cm file from the Sensor Calibration CD (see Step 1) to a USB disk.

b. Insert the USB into the robot controller and navigate to where the .cm file was saved. (MENU>>FILE>>File>>F5, [UTIL]>>Set Device>>USB Disk (UD1:)).

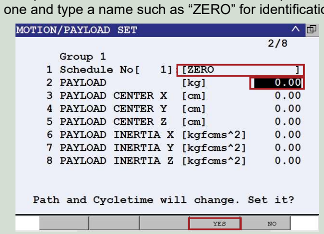
c. With the .cm file highlighted, press ENTER to execute the .cm file. Press F4, Yes, to execute the file.

d. Once the file is loaded, the sensor serial number should be shown in the DCS Collaborative robot screen. This number should match the collaborative sensor serial number that is physically on the robot.



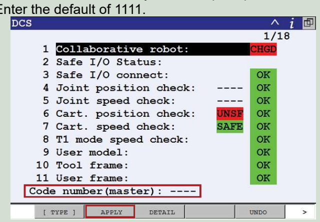
e. Turn the teach pendant ON/OFF switch to the ON position. Press MENU, cursor to NEXT, and press ENTER. Cursor to SYSTEM > Motion and press ENTER. On No. 1, press F3, DETAIL.

f. Cursor to Payload [kg], press 0, then press ENTER. You will prompted to answer the question shown below. Press F4, YES. Optionally, you can cursor up one and type a name such as "ZERO" for identification.



g. Press the PREV key and press F5, SETIND. Type 1 for the payload number, and press ENTER. This makes schedule 1 the active payload.

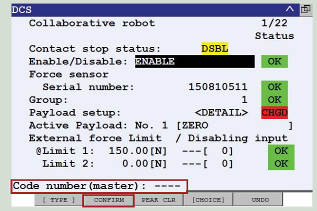
h. Press MENU, choose NEXT, cursor to SYSTEM >> DCS, and press ENTER. Press F2, APPLY, you will be prompted for the DCS code number. Enter the default of 1111.



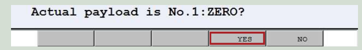
i. On the following screen, press F4, OK, to continue.

j. You must cycle power on the controller. Turn the Breaker off, wait 5 seconds, and then turn it back on. Alternatively, you can press the FCTN key, cursor to 0 NEXT, and press ENTER. Then cursor to 8 CYCLE POWER, press ENTER, and select YES.

k. After the robot has cycled power, the screen below will be displayed. Press F2, CONFIRM, you will be prompetd for the DCS code number. Enter the default of 1111.



l. You will be prompted with the current payload schedule. If No. 1 is displayed similar to below, confirm by pressing F4, YES. If another number is displayed press F5, NO, and repeat the procedures starting at D1e.



m. You will be prompted to check that no person or object is in contact with the robot arm. When verified, press F4, YES.

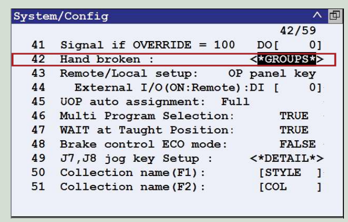
You should see "Payload confirmation success" at the bottom of the screen. Continue to the next step.

2. General settings

a. Press MENU, select 0 (Next), and select (SYSTEM), and then select CONFIG.

- Use the teach pendant arrow keys to move the cursor to the Hand Broken option, and press ENTER.

- Use the teach pendant function keys to disable the hand broken setting for each motion group. **NOTE:** It might already be disabled on your system.



b. Press MENU, select 0 (Next), and select (SYSTEM), and then select CONFIG.

- Use the teach pendant arrow keys to move the cursor to the Enable UI signals option.

- Use the teach pendant function keys to set the option to FALSE.

c. Ensure the Emergency Stop buttons on the teach pendant and controller are popped out. If not, twist the button clockwise to reset.

3. Press and hold a DEADMAN switch on the back of the teach pendant to its middle position. Press RESET on the teach pendant to clear alarms.

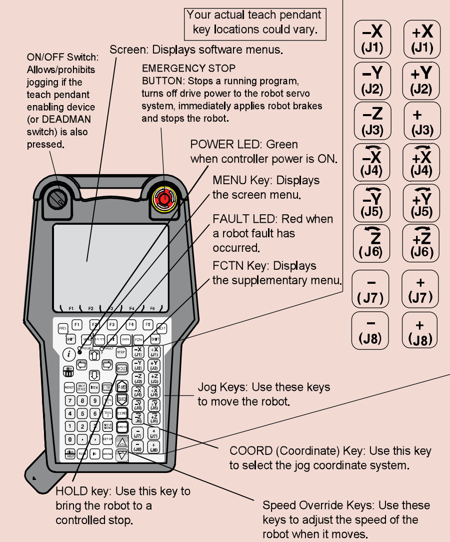
NOTE: To view alarms, press MENU, cursor to ALARM, and press ENTER. To toggle between alarm history and active alarms, press F3.

To move each joint axis of the robot, press and hold SHIFT and press each jog key one at a time and watch the robot joint move. Refer to the **Robot (mechanical unit) with joints indicated** diagram on Page 1 of this guide.

Try jogging each joint of the robot.

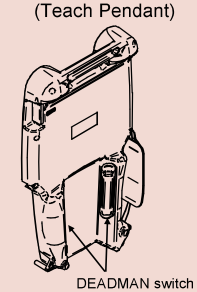
Jogging the robot is complete. You are now ready to perform software installation, if necessary. Refer to the *Controller-specific Software Installation* manual.

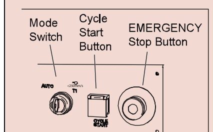
Reference



Refer to your *Application-Specific Setup and Operations Manual* for more information on each teach pendant key.

*i*Pendant (Teach Pendant)





Controller Operator Panel

For more information

To access controller, robot, or application-specific manuals, scan the **QR Code** to the right or go to **https://www.fanucamerica.com/**.

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